

Intelligent Molecule Finder

A Computational Chemistry Web Application:

Bridging Computer Science and Drug Discovery

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Problem Statement

The Challenge

Scientists and students need to quickly analyze molecular properties for drug discovery and research. However, existing professional tools are often prohibitively expensive, overly complex to install, or require deep chemistry expertise to operate effectively.

The Need

There is a critical need for an accessible, zero-setup, web-based tool that democratizes molecular analysis. The solution must provide instant visualization and property calculation without the steep learning curve of industrial software.

Project Objectives

- ✓ **Build a user-friendly web application** for instant molecular search and analysis.
- ✓ **Integrate with PubChem**, the world's largest chemical database, for robust data retrieval.
- ✓ **Implement functional group detection** covering 70+ chemical groups using substructure matching.
- ✓ **Calculate drug-likeness properties** based on pharmaceutical rules like Lipinski's Rule of Five.
- ✓ **Generate interactive 3D visualizations** to bring static chemical formulas to life.

Technology Stack



Python

Core programming language for backend logic and data processing.



Gradio

Web framework used for rapid UI development and deployment.



RDKit

Open-source cheminformatics software for molecular manipulation.



PubChem API

REST API source for retrieving data on 100M+ compounds.

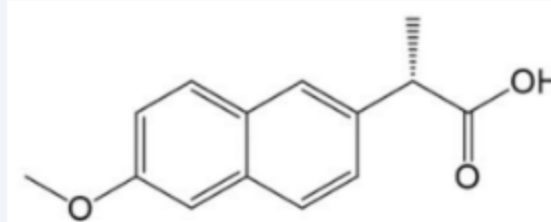
Core Concepts: What is a Molecule?

A Digital Representation of Matter

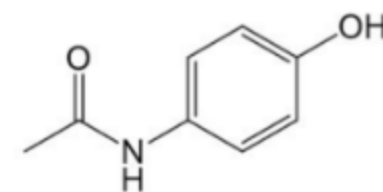
A molecule is a group of atoms bonded together, representing the smallest fundamental unit of a chemical compound.

Why Computers Care:

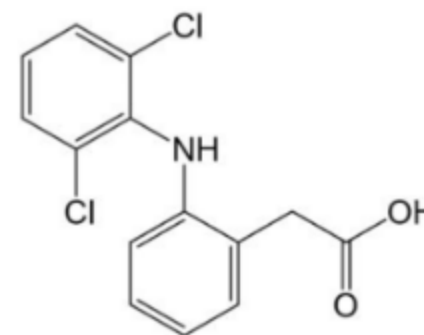
- **Drug Discovery:** Simulating interactions between drugs and proteins.
- **Materials Science:** Designing new batteries or polymers.
- **Environmental Analysis:** Tracking pollutants.



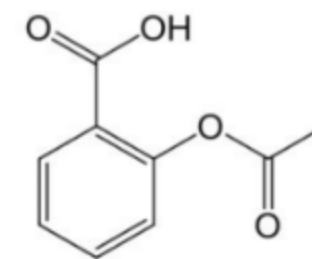
Naproxen



Paracetamol



Diclofenac



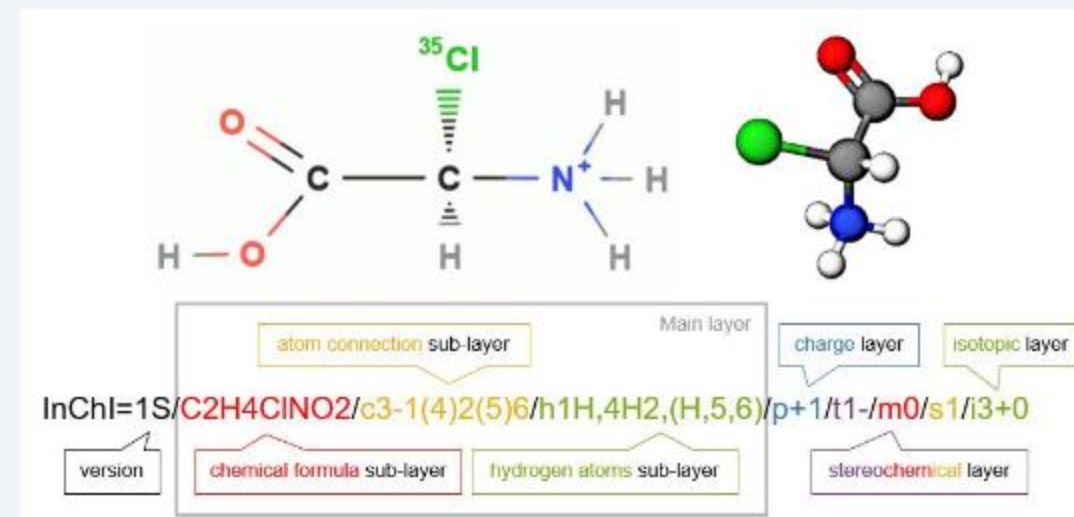
Aspirin

Chemical Representation Systems

How to Speak "Molecule"

Computers cannot "see" atoms, so we use string-based notations:

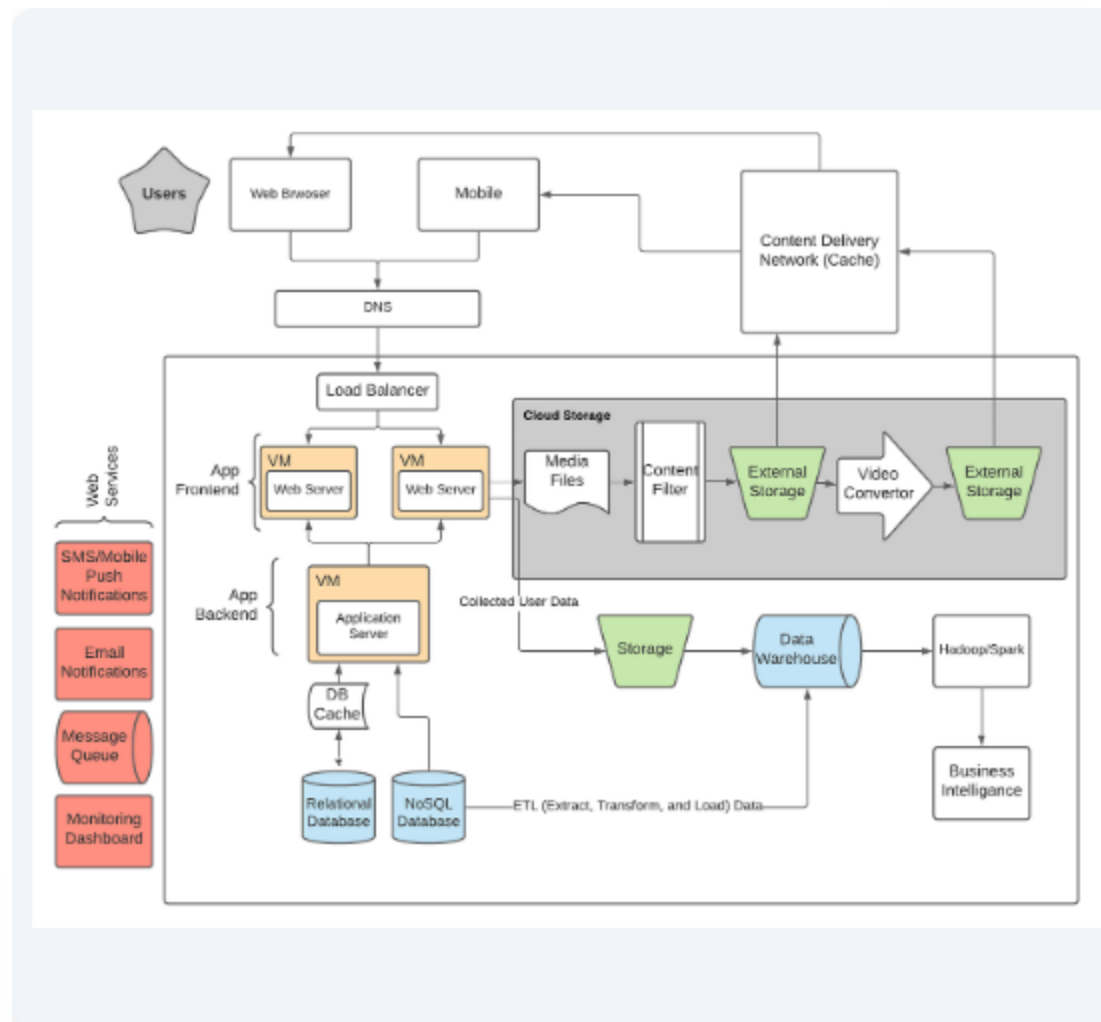
- **SMILES (Simplified Molecular Input Line Entry System):** Text string describing the structure (e.g., "CCO" for Ethanol).
- **InChI (International Chemical Identifier):** A standardized, unique string ensuring every molecule has one ID.
- **SMARTS:** A language for specifying substructures (used for pattern matching).



System Architecture

Data Flow

1. **User Input:** Enters molecule name (e.g., "Aspirin").
2. **API Query:** App requests data from PubChem.
3. **Data Processing:** RDKit converts SMILES to a Molecule Object.
4. **Analysis Engine:** Detects functional groups and calculates properties.
5. **3D Generation:** ETKDG + UFF force field embedding.
6. **Output:** Gradio renders the UI.



Component 1: PubChem Interface

REST API Integration

What: A module to fetch molecular data dynamically.

How: Sends HTTP GET requests to the PubChem PUG REST service.
Returns JSON data containing Canonical SMILES and properties.

```
import requests
def get_smiles(compound_name):
    url = f"https://pubchem.ncbi.nlm.nih.gov/rest/pug/compound/name/{compound_name}/property/CanonicalSMILES/JSON"
    response = requests.get(url)
    if response.status_code == 200:
        data = response.json()
        return data['PropertyTable']['Properties'][0]['CanonicalSMILES']
    return None
```

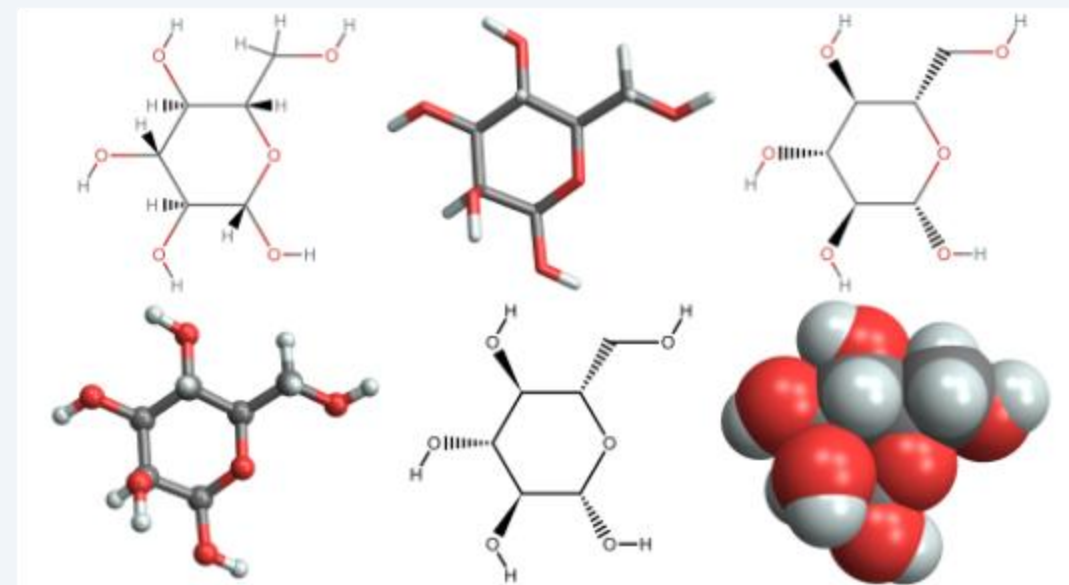

Component 2: Functional Group Detection

Substructure Matching

What: Identifying specific parts of molecules (like an Alcohol group or Benzene ring) that determine chemical behavior.

Method: Uses SMARTS pattern matching algorithms (Graph Isomorphism).

Coverage: The system checks for 70+ groups across 8 categories (Alcohols, Ketones, Aromatics, etc.).



Component 3: Property Calculation

Property	Definition	Relevance
Molecular Weight	Sum of atomic masses in the molecule.	Affects absorption and distribution in the body.
LogP	Lipophilicity (fat solubility).	Determines how well a drug crosses cell membranes.
Polar Surface Area (TPSA)	Surface sum of polar atoms (O, N, etc.).	Predicts transport properties and permeability.
H-Bond Donors/Acceptors	Number of H atoms available for bonding.	Crucial for binding to biological targets.

Drug-Likeness Assessment

Why Assess Drug-Likeness?

Not all molecules can be drugs. We use established pharmaceutical heuristics to filter viable candidates.

The application automatically evaluates molecules against these strict rules, providing a Pass/Fail report.

Rule Set	Criteria
Lipinski's Rule of 5	MW < 500, LogP < 5, H-Donors < 5, H-Acceptors < 10
Veber's Rules	Rotatable Bonds ≤ 10 , TPSA $\leq 140 \text{ \AA}^2$
Ghose Filter	Strict ranges for mass, refractivity, and atoms.

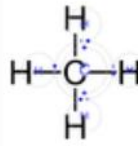
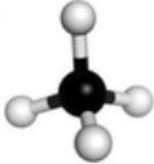
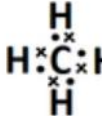
3D Structure Generation

From Flat Text to 3D Reality

Challenge: SMILES strings are 1D, but molecules exist in 3D space.

Solution:

- **ETKDG Algorithm:** Uses distance geometry and experimental torsion knowledge to guess coordinates.
- **UFF Optimization:** Applies a Universal Force Field to "relax" the molecule into a low-energy, realistic shape.

		
	Electrons are in fixed positions in the model, but in reality they constantly move, but are on average found between the two nuclei they bond.	Not the true shape of the molecule. It shows the bonds as 90° whereas they are 109.5° Shows a 3D shape with correct bond angles of 109.5°
Formula showing bonds	Shows which atoms are bonded together.	
Dot and cross diagram showing outer electrons.	Better for showing the angles of the bonds in a 2D way.	
Dot and cross diagram showing shells as circles	Shows which atoms are bonded together.	
Stick model 3D	Shows which atom the electrons came from originally	
Electrons are in fixed positions in the model, but in reality they constantly move, but are on average found between the two nuclei they bond.	In reality all electrons are identical.	
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	Shows which atom the electrons came from originally	

User Interface

Built with Gradio for simplicity and speed.

Key Components:

- Text Input for Molecule Name.
- Summary Data Table.
- Functional Group Detection List.
- Interactive 3D Viewer (py3Dmol).

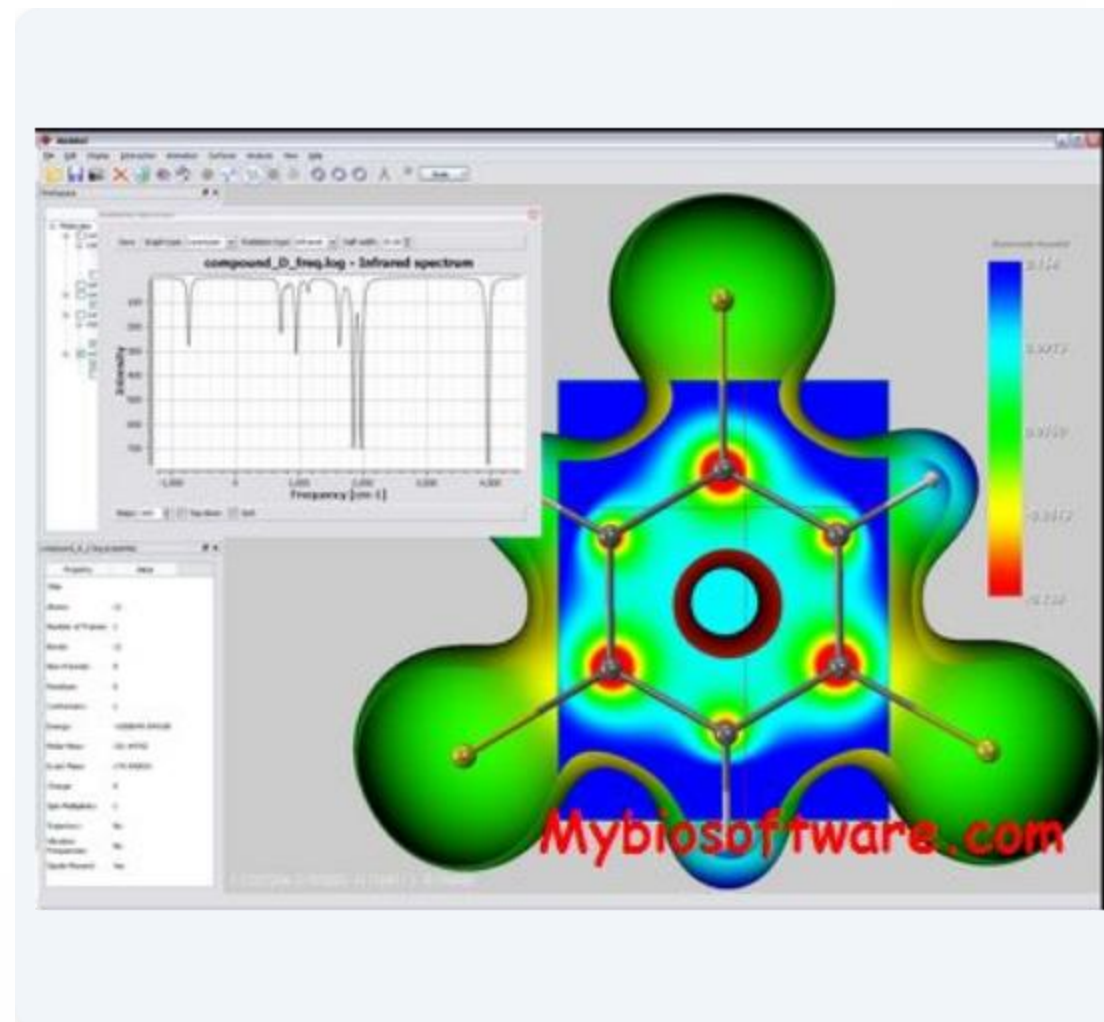
The clean design ensures usability for non-chemists.



Live Demonstration

Demo Scenarios

1. **Search for "Aspirin"**: Observe API retrieval speed.
2. **View Functional Groups**: Identify the Carboxylic Acid and Ester groups.
3. **Analyze Drug-Likeness**: Check if it passes Lipinski's rules (It does).
4. **Rotate 3D Structure**: Interact with the generated model in the browser.



Future Enhancements

- **Similarity Search:** Find existing drugs similar to the query molecule.
- **Batch Processing:** Upload a CSV of 1000s of molecules for bulk analysis.
- **Advanced Rendering:** Add surface, cartoon, and stick rendering styles.
- **Machine Learning:** Integrate toxicity prediction models using Scikit-Learn.
- **Export Feature:** Download analysis results as PDF or CSV reports.

Questions?

Thank you for your attention.